LoViF 2026 The First Challenge on Holistic Quality Assessment for 4D World Model (PhyScore)

Luo, Wei, Lu, Yiting, Li, Xin, Li, Haoran, Guan, Fengbin, Gao, Chen et al. | cs.CV

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Research Questions

- How can we evaluate world model-generated videos beyond perceptual quality to include physical plausibility, temporal coherence, and condition alignment?
- What metrics and architectures are most effective for localizing physical anomalies in generated 4D content?
- How do different design choices in holistic video assessment models trade off between accuracy, physics awareness, and computational efficiency?

Core Idea

Current evaluation of world model videos focuses only on visual quality, missing critical physical violations and temporal inconsistencies. This challenge introduces a holistic assessment framework that jointly evaluates four dimensions: video quality, physical realism, condition-video alignment, and temporal consistency, plus localizes anomaly timestamps. The key insight is that effective world model evaluation requires physics-aware representations and multi-dimensional scoring rather than appearance-only metrics.

Key Formula

Composite Evaluation Score

$$\text{Final Score} = 0.2 \times \text{IOU} + 0.4 \times \text{SRCC} + 0.4 \times \text{PLCC}$$

IOU measures anomaly timestamp localization accuracy, SRCC and PLCC measure correlation between predicted and ground truth scores across four quality dimensions (averaged). This formula balances temporal localization (20%) with overall quality prediction accuracy (80%) for holistic world model assessment.

Key Figure

Figure 3. Thumbnails and multidimensional scores of 10 generated videos in the training set: example 3

Table 1. Quantitative results on the PhyScore testing leaderboard (CodaBench). The best and second best values per column are highlighted in red and blue, respectively. For **ConditionVideoAlignment** (sorted ascending), **Runtime**, and **ExtraData**, lower is better as defined on the leaderboard. TimeStamp_IOU measures anomaly timestamp localization quality. The ranking index matches the numeric prefix of the corresponding team folder in this project.

Rank	Team	Final Score ↑	VideoQuality ↑	PhysicalRealism ↑	ConditionVideoAlignment ↓	Consistency ↑	TimeStamp_IOU ↑	Runtime [s] ↓	CPU[1]/GPU[0]	ExtraData[1]/No[0] ↓
1	SJTU_MM	0.532	0.457	0.497	0.582	0.519	0.607	12.0	0	1
2	IHNI	0.525	0.467	0.489	0.562	0.501	0.605	15.0	0	0
3	WDL	0.486	0.429	0.447	0.517	0.457	0.580	12.0	0	1
4	zhilong	0.484	0.478	0.417	0.528	0.460	0.536	12.0	0	1
5	DYTH	0.399	0.362	0.305	0.461	0.329	0.538	60.0	0	1

the best **VideoQuality** score, while DYTH obtains the lowest (best) **ConditionVideoAlignment** value as defined by the leaderboard sorting rule.

These results highlight key trade-offs in metric learning for world-model videos. Methods that emphasize physics-aware representations tend to benefit **PhysicalRealism** and anomaly timestamp localization, whereas specialized perceptual modeling can improve **VideoQuality**. In addition, runtime and extra-data usage further differentiate practical deployments, suggesting that effective holistic evaluators must balance accuracy, generalization, and efficiency.

3.1. Experimental Analysis

Drawing inspiration from recent short-form UGC video quality assessment challenges, we summarize several key findings and common practices from the submitted methods that lead to superior performance in holistic quality assessment for 4D world models.

1. Architectures and Feature Extractors: A majority of the participating teams utilize multi-branch architectures to capture different dimensions of video quality. Spatial features are frequently extracted using powerful backbones such as ConvNeXt or Swin Transformer, while temporal and motion distortions are captured via SlowFast networks or VideoMAE. Furthermore, the integration of Large Multi-Modality Models (LMMs) and foundational visual encoders has proven highly effective in aligning visual features with semantic and quality-aware representations, which is crucial for evaluating **ConditionVideoAlignment** and **VideoQuality**.

2. Physics-aware and Anomaly Localization Strategies: Unlike traditional UGC video quality assessment, evaluating videos generated by 4D world models requires specialized modeling for physical realism and spatio-temporal consistency. Methods that explicitly incorporate physics-aware representations or optical flow modules tend to achieve better anomaly timestamp localization (**TimeStamp_IOU**) and higher **PhysicalRealism** scores. This indicates that tracking temporal dynamics and physical constraints is indispensable for holistic world-model evaluation.

3. Ensemble Methods and Training Strategies: Consistent with the findings in existing VQA challenges, ensemble strategies are widely adopted by the top teams to boost prediction accuracy and stability. Model ensembles across different network architectures or fusing model weights from different training epochs significantly enhance the robustness of the final predictions. Additionally, training tricks such as Exponential Moving Average (EMA), rank-aware loss optimization, and advanced data augmentation strategies are commonly utilized to prevent overfitting and ensure stable model convergence.

4. Extra Data and Pre-training: The employment of extra training data and pre-trained weights is prevalent among the leading submissions. Leveraging models pre-trained on large-scale natural or synthetic video datasets provides a strong prior for general video quality assessment. Specifically, utilizing additional datasets helps the models generalize better to the complex and diverse distortions generated by world models, thereby yielding more accurate predictions across the four evaluated dimensions.

5. Trade-offs between Accuracy and Efficiency:

Table 1: Shows quantitative results comparing all submitted methods across the four evaluation dimensions plus runtime and data usage, revealing key trade-offs between physics awareness, perceptual quality, and efficiency.

Limitations

- Dataset limited to 1,554 videos from only 7 world models, potentially limiting generalization
- Subjective annotation process relies on only 4 trained annotators, introducing potential bias
- Physics categories focus primarily on dynamics, optics, and thermodynamics, missing other physical principles
- Composite scoring weights (0.2 IOU, 0.4 SRCC, 0.4 PLCC) are fixed rather than task-adaptive